Functions: Composite, Inverse & Logarithmic

Section 1: Inverses & Domain Restrictions

1 If $f(x) = \frac{2x-5}{3x+1}$, what is the value of $f^{-1}(1)$?

a. -6

b. -5

c. -4

d. -2

2 What is the domain of $f^{-1}(x)$ if $f(x) = \sqrt{3-2x}$?

a. $[0, \infty)$

b. $(-\infty, 1.5]$

c. $[1.5, \infty)$

d. All real numbers

3 Given $f(x) = x^2 - 4x + 7$ defined on the restricted domain $[2, \infty)$, what is the equation of $f^{-1}(x)$?

a. $f^{-1}(x) = 2 + \sqrt{x-3}$

b. $f^{-1}(x) = 2 - \sqrt{x-3}$

c. $f^{-1}(x) = \sqrt{x-7} + 2$

d. $f^{-1}(x) = 2 + \sqrt{x+3}$

Section 2: Composition & Decomposition of Functions

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4 If $f(x) = 3x - 2$ and $(f \circ g)(x) = 6x^2 + 15x + 4$, what is the expression for $g(x)$?

a. $2x^2 + 5x + 2$

b. $2x^2 + 5x - 2$

c. $3x^2 + 5x + 2$

d. $2x^2 + 3x + 4$

5 Let $f(x) = \frac{1}{x}$ and $g(x) = \sqrt{x - 2}$. What is the domain of $(f \circ g)(x)$?

a. $(2, \infty)$

b. $[2, \infty)$

c. $[0, \infty)$

d. $x \neq 2$

6 If $h(x) = (2x^2 - 1)^5$, identify inner function $g(x)$ and outer function $f(u)$ such that $h(x) = f(g(x))$.

a. $g(x) = 2x^2 - 1$ and $f(u) = u^5$

b. $g(x) = x^5$ and $f(u) = 2u^2 - 1$

c. $g(x) = 2x$ and $f(u) = (u - 1)^5$

d. $g(x) = 2x^2$ and $f(u) = u^5 - 1$

Section 3: Logarithmic Laws & Base Changes

7 Solve the logarithmic equation: $\log_4(x + 3) + \log_4(x - 3) = 2$.

a. 5

b. -5

c. 5 and -5

d. $\sqrt{19}$

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8 Using the change of base formula, evaluate $\log_8(32)$.

a. $\frac{5}{3}$

b. $\frac{3}{5}$

c. 4

d. $\frac{4}{3}$

9 If $\ln x + \ln y = a$ and $\ln x - \ln y = b$, what is x?

a. $e^{(a+b)/2}$

b. e^{a+b}

c. e^{ab}

d. $\frac{e^a + e^b}{2}$

Section 4: Exponential Models & Logarithmic Graphs

10 What is the vertical asymptote of the graph of $f(x) = 3\ln(2x - 8) + 1$?

a. $x = 4$

b. $x = 8$

c. $x = 0$

d. $y = 1$

11 Solve the exponential equation for t: $500e^{0.04t} = 2000$.

a. $25\ln 4$

b. $\frac{\ln 4}{0.04}$

c. $50\ln 2$

d. All of the above

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12 What is the x-intercept of the graph of $g(x) = \log_2 (x - 3) - 2$?

a. 7

b. 5

c. 4

d. 11

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